Controlled Evaluation of Class-Imbalance Mitigation Across Histopathology Patch Classification and WSI-Level MIL

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August 13, 2026

Abstract

Class imbalance is difficult to isolate in computational pathology because raw sample counts can hide two different shortages: few labelled patients or slides, and few informative patches within them. This benchmark therefore asks a controlled counterfactual question: how would the same prediction task change if one fixed training budget were distributed evenly across classes rather than concentrated in head classes? CAMELYON16, TCGA-UT, BRACS, and PANDA are studied in patch-classification and whole-slide multiple-instance-learning (MIL) regimes using frozen Virchow2 features. Patient- or case-disjoint partitions, natural validation and test cohorts, model families, and optimizer-update budgets remain fixed while only training support is redistributed among size-matched balanced, moderate, and severe conditions. Balanced accuracy and tail-group macro negative log likelihood measure discrimination and probability quality. Mitigation recovery is interpreted only after a prespecified gate establishes a meaningful imbalance-induced deficit. Exploratory analyses then relate damage and recovery to achieved severity, class difficulty, and independent support.

Contents

1 Introduction

Class imbalance is usually described by counting examples per class. In pathology, that count can be deceptive. Thousands of patches may come from only a few patients and therefore provide much less independent evidence than their raw number suggests. At whole-slide level, the opposite problem is explicit: a rare diagnosis may be represented by only a handful of independently labelled slides. Frequent classes then dominate the empirical objective, while rare classes may provide too little evidence for stable decision boundaries or reliable probabilities (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Comparing naturally imbalanced datasets cannot identify this effect by itself. Such datasets differ simultaneously in task difficulty, class separability, training volume, label granularity, and evaluation prevalence. The benchmark instead asks a narrower counterfactual question: *for the same task and the same total training budget, what changes when support is allocated evenly rather than concentrated in head classes?* A size-matched balanced cross-entropy (CE) model provides the reference. Validation and test cases remain natural and unchanged, so any balanced–imbalanced difference is tied to training allocation rather than to a different evaluation population.

This counterfactual creates a sequence of three research questions:

RQ1. Effect of imbalance. At fixed task, split-unit partition, encoder, training-set size, and evaluation set, how does increasing training imbalance affect tail-class discrimination and calibration in patch classification and WSI-level MIL?

RQ2. Mitigation effectiveness. Which mitigation methods recover performance lost specifically to training imbalance, and what are their calibration and computational costs?

RQ3. Predictors of damage and mitigation headroom. Can small exploratory models help explain why some controlled comparisons are more vulnerable than others? Specifically, are achieved imbalance severity and support–difficulty alignment associated with whether damage occurs, how large it is, and how much is recovered after damage? Separability, learnability, and support measures are examined as sensitivity predictors.

RQ1 first establishes whether a recoverable problem exists. RQ2 is evaluated only against that demonstrated deficit. RQ3 then asks why the size of the deficit and available headroom differ; it is an exploratory association analysis, not another method ranking. Dataset provenance and detailed method objectives remain in the companion reports (Queisler, 2026a,b). The next section turns this sequence into the benchmark's organizing logic.

2 Benchmark Logic

Interpretability depends on separating the intended intervention from everything else. The benchmark therefore freezes three layers of the comparison. Task definition fixes the data source, prediction target, and frozen representation. Evidence design fixes the split-unit partition and total controlled support. Model comparison fixes the natural validation and test cohorts and the training budget. Only class allocation within training support changes. Figure 1 follows that contrast from construction to interpretation.

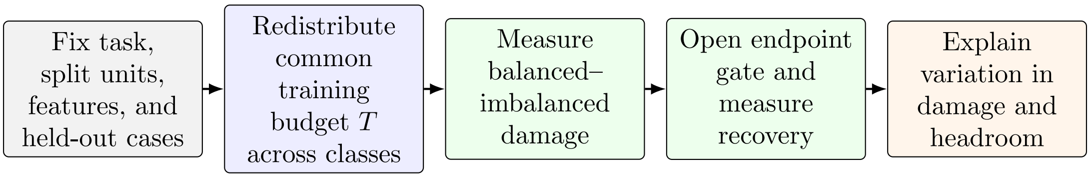

Figure 1: Logic of the benchmark. RQ1 concerns the damage contrast, RQ2 concerns gated recovery, and RQ3 concerns variation among those controlled comparisons.

One *comparison unit* is a dataset, prediction regime, imbalance severity, and tail assignment evaluated together. A *deficit gate* is a prespecified rule that asks whether imbalance caused enough damage, with sufficient precision, to make a recovery ratio interpretable. *Recovery* is the fraction of that demonstrated deficit regained by a mitigation method. These are analysis concepts, not model components.

Every quantity that this logic holds fixed is prespecified before the first definitive fit. Table 1 collects them with a pointer to the specification that derives each one, so the sections that follow can argue about the intervention rather than restate its settings.

Locked quantity	Value	Spec
Representation	Frozen Virchow2 features	A
Patch classifier	Two-layer MLP, 512 hidden units	A
Update budget	$U = 30\lceil T/B \rceil$ optimizer updates	D
Learning-rate envelope	Seven values, 1×10^{-5} to 1×10^{-2}	D.2
Support floor per class	≥ 10 split units and ≥ 20 slides	A.2
Contribution caps	$\leq 10\%$ per split unit; $\leq 5\%$ per slide	A.2
Severity targets	$\rho_{\text{target}} \in \{1, 10, 100\}$	A
Severity tolerance	Achieved $\rho \in [9, 11]$ and $[90, 110]$	A.2
Tail assignments	Native, difficulty-aligned, difficulty-reversed	A
Split-unit partitions	Three locked, patient- or case-disjoint	A
Initialization seeds	Two tuning; five disjoint confirmation	D
Discrimination gate	$D_{\text{BA}} \geq 0.02$, paired CI excludes 0	5.5
Calibration gate	$D_{\text{cal}} \geq 0.05$ nats, paired CI excludes 0	5.5
Bootstrap replicates	10,000, split-unit-clustered	B

Table 1: Quantities locked before the first definitive fit, with the section or appendix that specifies each. Only class allocation within the shared training support T varies across conditions.

This logic also explains the order of the report. Section 3 establishes what constitutes a prediction and an independent unit of evidence. Section 4 constructs the training intervention. Section 5 defines damage, recovery, and uncertainty. Section 6 explains heterogeneity, and Section 7 states the resulting claim boundaries. Exact allocation, training, metric, resampling, and model specifications are collected in the appendices.

3 Prediction Tasks and Evidence Units

Patch counts and independent evidence are not interchangeable. Patch classification predicts a label for one directly labelled image crop. WSI-level MIL predicts one label for a complete slide represented as a bag of patch embeddings; its patches are instances, not independently labelled examples. The terms *patch* and *tile* both denote a 256×256 crop, but *patch* is used throughout.

Table 2 defines the prediction example, target, and split unit in each dataset–regime. A *split unit* is the patient or participant where available and the biopsy or WSI otherwise. It is assigned to exactly one partition before any imbalance is constructed, and all slides from that unit follow the assignment.

Every eligible patch embedding enters the MIL model for its WSI, so bag size reflects the eligible tissue on that slide. The complete bag is identical across training conditions and methods. Eligibility is fixed before splitting and construction; Appendix A gives the exclusion and preprocessing rules.

Dataset	Patch-level example and label	MIL bag and bag label	Split unit
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	One WSI bag; metastasis-bearing (tumor) versus normal slide	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI's cancer type	One diagnostic-WSI bag; the same one of 31 eligible cancer types	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	One WSI bag; official seven-class WSI label defined by the most aggressive lesion detected in the slide	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	One biopsy-WSI bag; ordinal ISUP grade 0–5	Biopsy (one WSI)

Table 2: Dataset-specific definitions of patch-level examples and bag-level examples. Only TCGA-UT uses the same target in both regimes. CAMELYON16 and PANDA change diagnostic granularity, while BRACS changes from the annotated ROI subtype to the official most-aggressive-lesion WSI subtype. TCGA-UT's eligible target set omits one of the 32 annotated cancer types; Appendix A states which class and why.

Patch classification. Eligible patches are embedded once with frozen Virchow2. Concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). CAMELYON16 and PANDA use mask-derived patch targets, TCGA-UT patches inherit the source WSI's cancer type, and BRACS patches inherit the annotated ROI subtype. The common model is a two-layer multilayer perceptron (MLP); method-specific head changes are listed in Appendix D.

WSI-level MIL. A slide is one bag of frozen patch embeddings. CAMELYON16 bags use slide diagnosis, TCGA-UT bags use cancer type, PANDA bags use ordinal ISUP grade, and BRACS bags use the official seven-class WSI label defined by the most aggressive detected lesion. BRACS targets are read from official WSI metadata rather than reconstructed from ROI annotations. The common attention-MIL model maps instances to a 256-dimensional space, weights them by attention, and classifies their weighted average. Slide labels are the only supervision; patch labels, ROI labels, and masks do not supervise primary MIL runs (Brancati et al., 2022; Queisler, 2026a).

Comparative scope. Only TCGA-UT permits a direct patch-versus-WSI comparison. Both regimes predict the same cancer-type label within the same participant-disjoint cohort. In CAMELYON16, patch classification detects tumor tissue, whereas a WSI bag is classified as metastasis-bearing or normal. In PANDA, patch classification detects cancer tissue, whereas a WSI bag predicts the biopsy's ISUP grade. In BRACS, a patch predicts its ROI-specific subtype, whereas a WSI bag predicts the official most-aggressive-lesion subtype for the complete slide. The CAMELYON16, PANDA, and BRACS regimes are therefore separate dataset–target groups rather than paired measurements of one task.

With prediction and evidence units fixed, the next question is how to change class allocation without changing the task.

4 Constructing Controlled Imbalance

Controlled imbalance must alter class allocation without quietly changing data volume, held-out prevalence, or the pool of available evidence. Four training conditions establish the comparison: a full natural anchor, a size-matched balanced reference, a moderate imbalance targeting $\rho = 10$, and a severe imbalance targeting $\rho = 100$ where feasible. The natural anchor describes performance with all eligible training data and with the dataset's own class prevalence; because it is descriptive, only its CE reference is fitted. Only the three controlled conditions share total support T, carry the full method roster, and enter imbalance-deficit and recovery estimands.

4.1 What changes and what remains fixed

The construction asks a simple counterfactual question: how would the same prediction task perform if a fixed training budget were distributed evenly across classes rather than concentrated in a head class? Answering this question requires more than specifying an imbalance ratio. The comparison must first fix the split-unit partition, eligible examples, representation, evaluation cohort, and total controlled training support; only then can class allocation be changed. Figure 2 summarizes this separation between fixed design choices and the intended intervention.

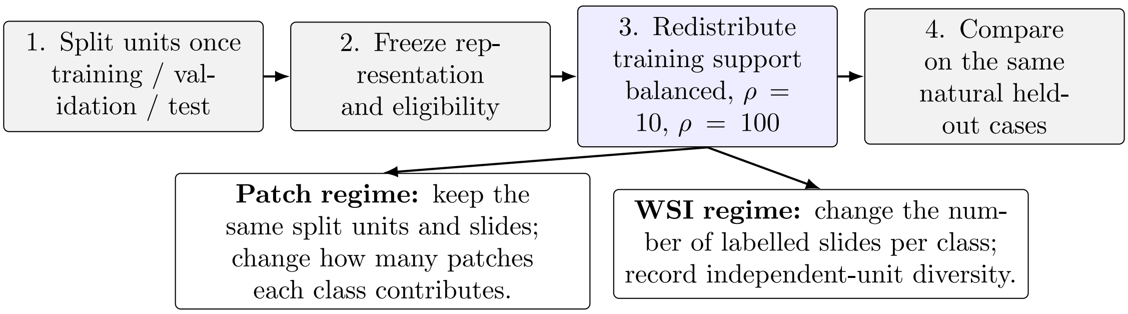

Figure 2: Controlled construction of the benchmark. Gray steps are fixed within a split; the blue step is the intended intervention. Patch imbalance redistributes patches within a common split-unit and slide pool, whereas WSI imbalance changes labelled slide support.

The balanced, moderate, and severe conditions contain the same total training support T: patches for patch classification and slides for MIL. The shared T is selected jointly with severity, not from size alone: construction retains the largest feasible total whose realized moderate and severe ratios lie within 10% of their targets, that is $\rho_{\text{achieved}} \in [9, 11]$ and $\rho_{\text{achieved}} \in [90, 110]$, for every locked tail assignment. Choosing T from size alone could flatten the intended contrast, and this tolerance keeps both controlled conditions recognizably close to their nominal severities rather than only as separated as the data allow. Appendix A specifies the tolerance search and its feasibility rules.

The target ratio $\rho_{\text{target}} \in \{1, 10, 100\}$ specifies the intended intervention, but integer allocation and finite inventories can change the realized class distribution. The achieved ratio is therefore recomputed from every manifest. For realized class supports $N_1, \ldots, N_K$, it is

$$\rho_{\text{achieved}} = \frac{N_{\max}}{N_{\min}}. \tag{1}$$

This quantity records the extreme head–tail contrast. The complementary normalized-entropy deficit

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^{K} p_c \log p_c}{\log K}, \qquad p_c = \frac{N_c}{\sum_j N_j}. \tag{2}$$

uses all classes and is normalized for class count: it is zero for a balanced allocation and increases as support becomes concentrated. Both quantities are computed separately at patch and slide level from each realized training manifest. Per-class split-unit, slide, and patch counts are reported alongside them, because neither scalar reveals whether many patches originate from only a few cases.

Split-unit-disjoint training, natural-validation, and natural-test partitions are created first. Imbalance manipulation is confined to training support. Within a split, every condition uses exactly the same validation and test cases and eligible held-out examples. No held-out example is deleted, duplicated, or reweighted according to the training severity. Patch-level evaluation uses every eligible held-out patch, and slide-level evaluation uses every eligible held-out slide. Consequently, recall is estimated from all available held-out cases and prevalence-dependent precision and macro F1 are compared at one fixed test prevalence.

The validation cohort also retains its natural distribution. It is used for hyperparameter and checkpoint selection but never resized to resemble a training condition. Training, validation, and test manifests are versioned with split-unit identifiers, class counts, construction seed, and content hashes before model fitting.

4.2 Independent support and feasibility

A patch is a training example, but thousands of patches from one patient do not constitute thousands of independent observations. *Independent support* therefore means the distinct split units and slides underlying a class. Patch conditions draw from fixed per-class patient and slide pools, whereas MIL imbalance necessarily changes the number of labelled slides and thus independent-unit diversity.

A *support pilot* determines how little independent tail evidence remains defensible before definitive comparisons are built. The definitive floor is the larger of that empirical level and the design floor implied by the contribution caps (Table 1). Conditions that cannot satisfy these floors are weakened or excluded rather than rescued by changing validation or test data.

4.3 Severity and tail placement

Severity controls how the shared total T is distributed. Binary tasks have one head and one tail; multiclass tasks place intermediate supports at equal intervals on a logarithmic scale. Requested and achieved ratios are both reported because finite inventories and integer counts can weaken the realized contrast. A condition that collapses to the balanced allocation is recorded as degenerate and never pooled with genuine imbalanced conditions.

Support rank must also be separated from intrinsic class difficulty. Native, difficulty-aligned, and difficulty-reversed *tail assignments* place semantic classes at different points in the same support profile. Aligned assignments compound difficulty with scarcity; reversed assignments compensate difficult classes with greater support. Duplicated assignments are omitted transparently. Appendix A gives contribution caps, pilot quotas, ordering rules, allocation equations, infeasibility handling, and seed families.

These safeguards define the evidence available to every model. The next section asks how matched models turn that evidence into damage and recovery estimates.

5 Measuring Damage and Recovery

RQ1 and RQ2 are answered in a deliberately asymmetric sequence. Balanced CE is compared with imbalanced CE first. Only when that contrast shows a meaningful deficit on discrimination or calibration is a mitigation method interpreted in terms of recovery on the affected endpoint.

For intuition, suppose balanced CE attains balanced accuracy 0.72, imbalanced CE attains 0.62, and a mitigation method attains 0.68. Imbalance caused a 0.10 deficit and the method regained 0.06, or 60% of the demonstrated damage. This normalization distinguishes repair of an imbalance effect from a generic method improvement. Exact estimands and gates appear after the endpoints they use.

5.1 Methods compared under imbalance

Table 3 lists only the methods evaluated in this benchmark. CE and MIL CE are the size-matched no-mitigation references. The roster is fitted in the balanced, moderate, and severe conditions, which alone supply the paired contrast a mitigation claim requires; the natural anchor carries the CE reference only. Method objectives, tested adaptations, ablations, defaults, and fidelity qualifications are defined exclusively in the companion methods report (Queisler, 2026b).

Family	Tested methods	Regime
No-mitigation reference	CE; MIL CE	Patch; WSI
Data-level sampling and augmentation	Balanced sampling; RankMix-inspired feature mixing	Both for sampling; WSI for RankMix-inspired
Losses and prior correction	Weighted CE; focal loss; train-time and post-hoc logit adjustment; CFAL	Both, except CFAL patch only
Hybrid sampling–loss	CE + soft F1; CE + soft MCC; OKO set learning; MDE-inspired dual-expert MIL	Patch for CE hybrids and OKO; WSI for MDE-inspired
Representation and architecture	cRT; SC-MIL	Both for cRT; WSI for SC-MIL

Table 3: Prespecified method roster, using the same intervention-level taxonomy as the companion methods report. “Both” means the patch and WSI implementations defined there, not a comparison of their raw scores.

5.2 Matched training and selection

All matched methods retain the frozen encoder, model family, optimizer family, batch definition, augmentation, numerical precision, and optimizer-update budget within a regime (Table 1). Hyperparameters and checkpoints are selected only on natural validation data, and are locked before confirmation fitting and the one-time test evaluation. Balanced CE, imbalanced CE, and mitigation runs share split, tail assignment, confirmation-seed index, and held-out cases. Appendix D gives update formulas, adaptive grids, tie-breakers, exposure accounting, failure rules, and cost reporting.

Figure 3 places these steps in execution order. It also separates the validation search that is self-contained from the phase that cannot start until a condition’s CE search has resolved, which is what fixes the order in which runs may be launched.

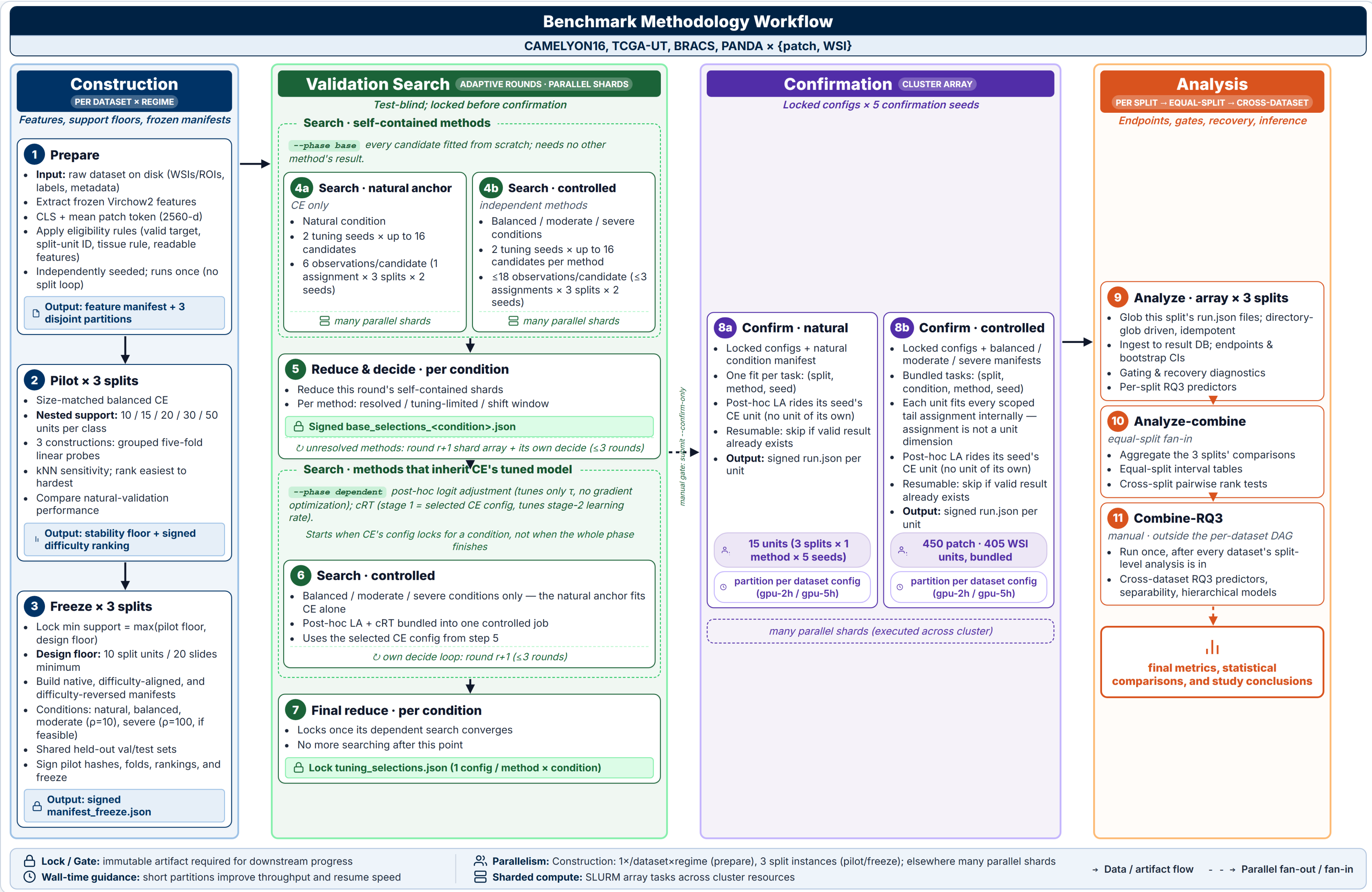

Figure 3: Operational workflow of the controlled class-imbalance benchmark.

This separation makes confirmation runs comparable without pretending that equal optimizer updates imply equal data exposure. With models fixed, evaluation can ask which endpoint was damaged and how much of that damage was recovered.

5.3 Discrimination endpoints and aggregation units

The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Overall accuracy is descriptive. Patch predictions are summarized in two ways: patch-micro estimates treat eligible patches as predictions while uncertainty remains split-unit-clustered, and slide/case-macro estimates first aggregate the metric contribution within each slide or split unit so that heavily tiled slides do not dominate. WSI outcomes are slide-level with patient clustering where patients contribute multiple slides and WSI/case clustering otherwise.

5.4 Ordinal error and probability quality

PANDA WSI prediction is the only ordinal task. Quadratic-weighted kappa distinguishes near from distant grade errors relative to the observed marginals, while ordinal mean absolute error reports grades missed per biopsy. These secondary outcomes complement rather than replace balanced accuracy.

Probability-quality metrics ask whether the complete predicted distribution is useful, not only whether its largest entry names the correct class. Tail-group macro negative log likelihood (NLL) is the co-primary calibration endpoint because it gives each tail class equal weight and penalizes confident errors. Natural-prevalence NLL, macro NLL, multiclass Brier score, expected calibration error (ECE), and reliability diagrams provide secondary views. ECE uses 10 fixed equal-width bins and remains secondary because small classwise samples make bin estimates unstable.

Discrimination uses balanced-decision logits, whereas calibration uses target-prior-corrected probabilities before temperature scaling. Temperature-scaled outputs are secondary and never enter a recovery gate. Appendix B defines PANDA's ordinal metrics, every probability-quality equation, clipping at 10^{-12}, score variants, and fixed-bin ECE.

5.5 From imbalance deficit to mitigation recovery

Recovery is meaningful only when imbalance has first caused a measurable loss. The analysis therefore proceeds in two stages: compare size-matched balanced and imbalanced CE, then evaluate mitigation only on an endpoint that shows sufficiently large and statistically supported harm. One *comparison unit* is a fixed dataset, prediction regime, imbalance severity, and tail assignment. A *gate* is the prespecified eligibility rule that routes such a unit to the relevant recovery analysis; it is not a learned model component.

For a higher-is-better metric M, define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$\begin{aligned} D_M &= M(\text{balanced CE}) - M(\text{imbalanced CE}), \\ R_M &= \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \end{aligned} \tag{3}$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T, methods that only reweight or recalibrate the imbalanced manifest cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M: it has no size-matched balanced counterpart, so neither a deficit nor a recovery fraction is defined there. Fitting mitigation methods on it would therefore yield differences confounded by training volume as well as allocation, and the anchor is restricted to its CE reference for that reason.

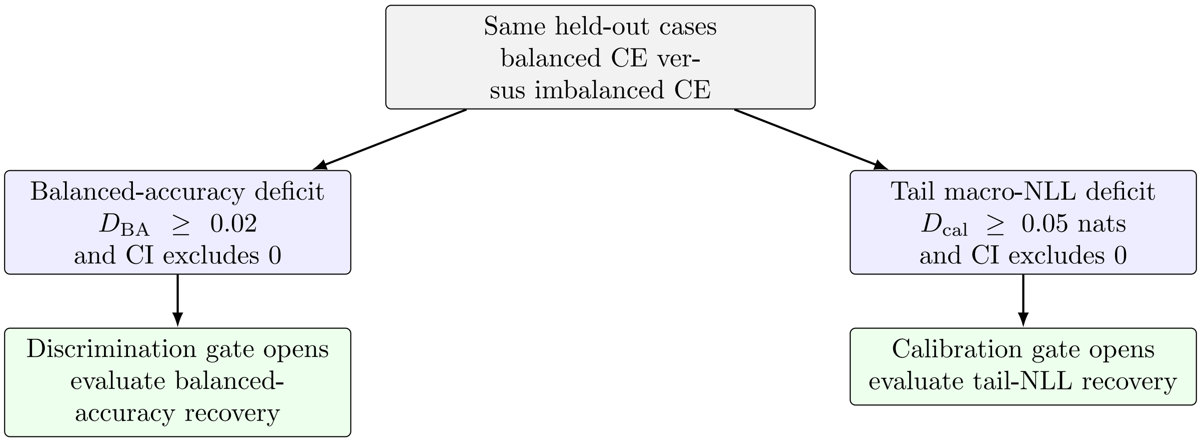

Figure 4: Routing from an imbalance deficit to mitigation analysis. A comparison unit can enter through either gate or both. If neither rule is met, its CE deficits are still reported, but a recovery ratio is not interpreted. CI denotes the paired 95% confidence interval.

Figure 4 shows the two routes. The *discrimination gate* opens when the CE balanced-accuracy deficit is at least 0.02 and its paired 95% confidence interval excludes zero. The *calibration gate* opens when the CE tail-group macro-NLL deficit, $\mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$, is at least 0.05 nats with a paired 95% confidence interval excluding zero. Both rules are computed from CE runs on the same held-out cases before any mitigation method is compared. A comparison unit entering through the discrimination gate is evaluated for balanced-accuracy recovery; one entering through the calibration gate is evaluated for tail-calibration recovery; it may enter through both.

The two routes preserve an important distinction. Strong frozen features may leave balanced accuracy nearly intact while imbalance still biases tail probabilities, in which case a single discrimination rule would exclude methods designed to correct class priors. Conversely, the gates avoid unstable recovery ratios when no meaningful damage exists to recover. All CE deficits on both axes remain reported, including units that fail both rules. For the lower-is-better calibration axis, the sign-corrected deficit $D_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$ is the denominator and the method's tail-NLL improvement over imbalanced CE is the numerator, so Equation (3) retains the same interpretation. Recovery for other secondary metrics is descriptive and does not reopen a comparison unit that fails both gates.

5.6 Uncertainty from independent units

Uncertainty follows patients or cases rather than patches. A split-unit-clustered Bayesian bootstrap draws one positive Dirichlet weight per unit and carries that weight to all of its slides, patches, split appearances, and matched methods. Balanced CE, imbalanced CE, and mitigation runs receive the same unit weights and paired confirmation-seed draws; the complete deficit and recovery fraction are recomputed in every replicate.

A label-only preflight fixes 10,000 resamples before predictions exist and checks class-cell representation, metric computability, paired weights, and weight variation. Kish effective counts and largest-weight shares identify cells dominated by too little independent evidence. Three split partitions are averaged with equal weight and are not treated as independent cohorts; five model initializations measure algorithmic variation, not five additional patient samples. Exact diagnostics, descriptive-only thresholds, and percentile-interval construction appear in Appendix B.

5.7 Paired testing and multiplicity control

Confidence intervals and raw effect sizes are the main evidence. Hypothesis tests provide a supporting check for the prespecified primary methods. On the discrimination axis, the tested effect is the balanced-accuracy improvement over imbalanced CE. On the calibration axis, it is the reduction in tail-group macro NLL, oriented so that positive values indicate improvement. Recovery fractions are normalized effect summaries, not test statistics.

The raw two-sided p-value comes from a paired split-unit-block permutation test. This test represents the null hypothesis of no method difference by repeatedly swapping the method and imbalanced-CE labels within a split unit. The same swap is used for all of that unit's slides, patches, split appearances, and matched confirmation seeds, preserving the paired design. All possible swaps are enumerated when feasible. Otherwise, 100,000 random swaps are used with a plus-one correction; this gives a minimum attainable Monte Carlo p-value of approximately 10^{-5} and avoids reporting a simulated value of zero.

A *test family* is the set of related hypotheses interpreted together. Testing many hypotheses increases the chance that at least one small p-value occurs by chance. Holm adjustment controls this family-wise error by ordering the raw p-values from smallest to largest, applying the strongest correction to the smallest value, and then proceeding stepwise with less stringent corrections. It provides this control without assuming that the tests are independent.

The confirmatory family contains weighted CE, focal loss, train-time logit adjustment, and balanced sampling. These four primary methods directly target tail support or class priors and are shared across both prediction regimes. Within one dataset and regime, the family includes every gate-passing combination of primary method, gate, severity, and tail assignment; severities are not split into separate families. Holm adjustment is applied across the testable comparisons, while gated-out comparisons are recorded as not tested. All remaining methods and all secondary classwise, calibration, and cost analyses are exploratory. They retain the same effect estimates and confidence intervals but remain outside the confirmatory correction.

6 Explaining Heterogeneity

Why do some controlled comparisons show more damage or recovery than others? RQ1 and RQ2 estimate matched effects within a dataset and task. RQ3 then takes one step back and relates prespecified properties of each comparison to those effects. It neither predicts patient outcomes nor creates another method ranking. For damage occurrence and magnitude, one observation is a dataset, prediction regime, severity, and tail assignment. Recovery observations additionally identify the method and gate.

Outcomes to explain. The analysis is divided into three linked questions rather than one opaque combined model:

1. **Damage occurrence:** did the comparison pass either prespecified deficit gate? Every controlled comparison contributes to this binary analysis.

2. **Damage magnitude:** how large was the signed balanced-accuracy deficit? Every comparison contributes, including those with no damage or an apparent improvement under

imbalance.

3. **Mitigation headroom:** how much of the demonstrated deficit was recovered? Only gated comparisons contribute because recovery has no stable interpretation when there is no deficit to recover.

These three outcomes correspond directly to the sequence introduced in RQ3: occurrence, magnitude, and conditional recovery.

Candidate explanations. Only quantities available without using test outcomes are eligible as predictors. Table 4 separates the two primary predictors from the variables used only in sensitivity analyses.

Predictor	Question it represents	Model role
Achieved imbalance ratio	How extreme was the realized head–tail allocation?	Primary, log-transformed
Support–difficulty alignment $A = \mathrm{corr}(\log N_c, d_c)$	Does greater allocated support go to more difficult classes ($A > 0$), or does scarcity compound difficulty ($A < 0$)?	Primary
Intrinsic separability	How distinguishable are the classes in a common balanced frozen-feature reference, before condition-specific support is considered?	One-predictor sensitivity
Condition-specific learnability	What remains learnable from the exact controlled training manifest?	One-predictor sensitivity
Minimum independent support	How many distinct split units or slides support the scarcest class?	One-predictor sensitivity
Effective patch support	How much patch support remains after accounting for within-case redundancy?	Patch-only sensitivity
Prediction regime	Does the comparison use patch classification or WSI-level MIL?	One-predictor sensitivity

Table 4: RQ3 predictors and their prespecified roles. Sensitivity predictors are fitted one at a time rather than added to the primary model.

Intrinsic separability asks whether frozen Virchow2 features distinguish classes in a common balanced reference. Condition-specific learnability asks what remains learnable from the exact controlled manifest. Both are measured on natural validation data without test outcomes. Minimum independent support and a patch-only redundancy-adjusted effective support are sensitivity predictors rather than replacements for patient or slide counts.

One deliberately small model is fitted for each outcome. Every primary model contains only log achieved imbalance ratio and support–difficulty alignment, and each remaining predictor enters a separate one-predictor sensitivity model rather than being added to that primary model. Weak Gaussian priors and leave-one-group-out checks stabilize interpretation without claiming predictive generalization. Appendix C specifies probes, cross-fitting, ICC, effective support, model forms, MAP fitting, and sensitivity analyses.

RQ3 can therefore suggest which combinations of severity, difficulty, and support deserve follow-up, but it cannot turn the small set of dataset–target groups into general predictive evidence. The final section makes this and the benchmark's other interpretation limits explicit.

7 Interpretation and Limitations

The design supports deliberately narrow claims. Its strongest claim concerns effects of the training distribution under one frozen Virchow2 representation. It does not establish how end-to-end encoder fine-tuning would respond, whether another foundation representation would produce the same deficits, or whether constructed tails reproduce biological prevalence and referral pathways. The shared total T isolates allocation by discarding some available examples;

the full natural condition is therefore retained as a descriptive CE anchor, which places the controlled results on the scale of the complete training partition and of the dataset's own class prevalence. Because no mitigation method is fitted on that anchor, the benchmark reports no method comparison under natural prevalence, and its method claims are confined to the size-matched controlled conditions.

Method comparisons also identify methods, not universal mechanisms. Some interventions change sampling, objectives, representations, or architecture together. Equal optimizer updates do not imply equal information exposure: RankMix trains teacher and student, MDE consumes natural and balanced minibatches, and post-hoc logit adjustment performs no gradient optimization. MIL bags likewise vary with eligible tissue area. Processed examples, unique exposures, accelerator time, memory, and model size are reported rather than treated as equal.

Independent support limits inference more strongly than raw patch count. BRACS WSI has a scarcest-class slide ceiling that caps achieved severity near 1.35, and one split collapses moderate and severe allocations to the balanced manifest. It is therefore descriptive rather than confirmatory. BRACS patch classification is also descriptive because its patch-micro Kish effective support remains below the prespecified threshold under patient-level bootstrap weights. These exclusions expose evidence limits rather than hide inconvenient results.

Finally, calibration is conditional on the natural held-out prevalence and cannot be transported to another deployment population without explicit prior-shift analysis. Within these boundaries, RQ1 estimates damage caused by controlled training allocation, RQ2 estimates recovery only where that damage is established, and RQ3 remains a hypothesis-generating explanation of between-comparison variation.

A Construction and Support Specification

This appendix records dataset eligibility, hierarchical sampling, support floors, allocation, and seed rules. These details implement the controlled contrast introduced in Section 4; they do not add further estimands.

A.1 Eligibility and representation

Before examples are selected, each split unit is assigned to exactly one training, validation, or test partition, and all slides from that unit follow its assignment. A sample is excluded if it lacks a valid target or split-unit identifier, yields no patch under the dataset-specific tissue rule, lacks an annotation required for its patch label, or lacks readable frozen features. The same eligibility rules are applied before every imbalance condition.

TCGA-UT further excludes one of its 32 annotated cancer types, Diffuse Large B-cell Lymphoma, from the eligible target set. In every locked split this class contributes 20 participants and 20 slides, one slide per participant, so the 10% split-unit and 5% slide patch contribution caps (Section A.2) admit essentially no count above the definitive floor: the class's set of cap-feasible patch counts covers only a few percent of the range between that floor and its native support, with the first infeasible count one unit above the floor. No shared total is therefore both tolerance-feasible and cap-feasible for this class, independent of severity or tail assignment. The remaining 31 cancer types are eligible in both prediction regimes and satisfy the tolerance search below.

For patch classification, all eligible patches enter the pool before condition-specific subsampling. The common classifier is a two-layer MLP with a 512-unit hidden representation, ReLU activation, dropout, and a linear class output. CFAL replaces the linear output with a prototype-based affinity head; OKO adds a training-only auxiliary head discarded at evaluation.

For WSI-level MIL, every eligible embedding enters its slide bag. BRACS WSIs are sampled at 20x into non-overlapping 256×256 patches and use at least 10% Otsu foreground, grayscale standard deviation of at least 8, Canny edge evidence, and at least two tissue neighbours. SC-MIL adds a training projection head, while MDE-inspired MIL replaces the single classifier with two expert heads. Dataset provenance and complete preprocessing definitions remain in the companion dataset report (Queisler, 2026a).

A.2 Independent evidence and support floors

The independent-support rule of Section 4 is realized differently in the two regimes. For patch classification, a fixed per-class split-unit pool and, within it, a fixed slide pool are made available to the balanced and all imbalanced conditions: every condition draws its patches only from this pool. For MIL, a slide is itself the labelled example, so reducing slide support necessarily reduces independent support, and MIL contrasts are interpreted as controlled slide-support imbalance including its induced change in independent-unit diversity. Table 5 states the selection order and caps.

The condition with the largest patch allocation for a class covers the fixed pool in full, so no unit in it goes unused by every condition together. A smaller condition for the same class, in particular a severe tail count well below that maximum, may leave some pool units untouched; this is the intended shape of an imbalanced allocation, not a violation of the shared pool.

The smallest defensible tail support is estimated before the definitive comparison with a *support pilot*. These balanced CE runs ask when additional independent training units cease to materially improve natural-validation performance; they do not tune a hyperparameter or contribute observations to the definitive analysis. In the patch regime, one pilot unit is a split unit, or a slide where split unit and slide coincide. Each selected unit contributes the same fixed quota q_{patch} of patches, distributed as evenly as possible over its eligible slides. The quota is the

Regime	Selection order	Contribution caps	What imbalance changes
Patch	Split units, then slides within units, then patches; deterministic round-robin before any unit is revisited	$\leq 10\%$ of a class's patch allocation per split unit; $\leq 5\%$ per slide	Patches per class; the available split-unit and slide pools are unchanged
MIL	Split units before additional slides from the same unit; each slide contributes once	$\leq 10\%$ of a class's slides per split unit	Labelled slide support, and therefore independent-unit diversity

Table 5: Hierarchical selection and contribution caps by prediction regime. Every manifest reports per-class split-unit, slide, and patch counts, maximum contributions, and the fraction of the eligible independent pool retained.

largest value the patient and slide contribution caps admit at every candidate level and every construction ordering, held fixed so increasing a pilot level adds independent units rather than patches per unit. A caps-admissible quota is necessary but not sufficient: a unit cannot supply more patches than it holds, so a unit is eligible for a candidate level only if its own inventory meets the candidate quota, and the quota at a given level is set from that level's order statistic of per-unit inventory over the eligible units rather than from the scarcest unit in one seeded ordering. A candidate level that no quota can satisfy is dropped from the curve rather than reusing a looser quota from a larger level. In the MIL regime, one pilot unit is a labelled slide represented by all its eligible instances.

Nested pilots compare $10, 15, 20, 30$, and 50 units per class, or all available units when the scarcest class has fewer. They are repeated for three independently seeded *construction orderings*. An ordering is a fixed sequence of eligible units; its prefix of length m is simply the first m units in that sequence. Thus the 15-unit pilot retains the same first 10 units as the 10-unit pilot and adds the next five. At lower levels of the hierarchy, every split unit has a fixed slide order and every slide has a fixed patch order. Figure 5 illustrates this nesting.

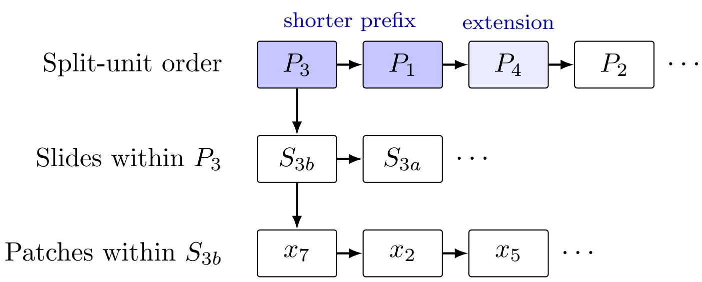

Figure 5: Hierarchy behind one construction ordering. A seed fixes the split-unit sequence and the local slide and patch sequences. The darker unit boxes form a shorter prefix; the lighter third box is added, without replacing earlier units, when the prefix is extended.

Because every larger pilot level extends the preceding prefix, adjacent levels are paired within an ordering and evaluated on the same natural-validation cohort. The smallest level whose next increment changes mean balanced accuracy by less than 0.01 and every class recall by less than 0.02 in all three orderings is the *stability floor*; if none qualifies, the largest candidate is used.[1]

The definitive minimum is the larger of this empirical stability floor and a prespecified design floor. Each class must contain at least 10 split units and 20 slides, or 20 slides when split unit and slide coincide, and any class-aware batch or set must contain at least two distinct same-class

[1]No pilot level is exempt from the caps. A level is dropped from the curve when no quota satisfies both caps for every construction seed—e.g., the 5% slide cap needs at least 20 contributing slides, so a class whose split units mostly contribute a single slide can leave levels below 20 infeasible. Dropped levels are recorded in the frozen pilot report.

units. The 10-unit and 20-slide values are not estimated performance thresholds: they are the smallest counts compatible with the 10% split-unit and 5% slide contribution caps, respectively. The same 20-slide lower bound is retained for MIL to keep slide-based method comparisons outside a five-example regime even though MIL has no separate 5% slide cap. A requested severity is reduced to the largest allocation satisfying both floors; a dataset–regime is excluded if even its balanced condition fails them. Pilot quotas, curves, seeds, and resulting floors are frozen and reported with the definitive manifests.

Within these floors, one shared total T is fixed for the balanced, moderate, and severe conditions of a split. Construction scores every integer total between the independent-support floor and the largest total any locked tail assignment's head class can support at $\rho_{\text{target}} = 100$, keeping only totals whose balanced allocation and whose every locked assignment's moderate and severe allocation satisfy the patch or slide contribution caps. Among these cap-feasible totals, the *tolerance-feasible* set is those whose achieved moderate ratio falls in $[9, 11]$ and whose achieved severe ratio falls in $[90, 110]$ simultaneously, for every locked assignment. The largest tolerance-feasible total is checked against the true fixed-pool construction — the same sampling procedure the definitive manifests use — starting from the largest candidate and moving to the next largest until one is confirmed; that confirmed total is T. A dataset–regime with no tolerance-feasible total retains its largest independent-support- and cap-feasible total instead, subject to the infeasibility fallback and degeneracy rule below.

Before assigning support ranks, the largest balanced pilot level is used to measure class difficulty. For each of the three pilot constructions, a class-balanced linear probe makes five-fold out-of-fold predictions with case/patient-disjoint `StratifiedGroupKFold` partitions. Class difficulty is $d_c = 1 - \overline{\text{recall}}_c$, where recall is averaged across the three constructions; five-neighbour kNN recall is retained as sensitivity evidence. Pilot hashes, folds, classwise scores, and the easiest-to-hardest ranking are signed. Equal scores retain native class order.

A.3 Severity profiles and tail assignments

A binary task needs only a head and a tail count, but a multiclass task also needs a rule for the classes between those extremes. The exponential construction places the K class supports at equal intervals on a logarithmic scale. For a locked order from head ($i = 0$) to tail ($i = K - 1$), it assigns weights

$$w_i(\rho_{\text{target}}) = \rho_{\text{target}}^{-i/(K-1)}, \qquad n_i^* = T \frac{w_i(\rho_{\text{target}})}{\sum_{j=0}^{K-1} w_j(\rho_{\text{target}})}, \quad i = 0, \ldots, K-1. \tag{4}$$

so the head-to-tail ratio is exactly ρ_{target} before integer rounding. For example, with four classes and $\rho_{\text{target}} = 100$, the unnormalized weights are approximately $1, 0.215, 0.046$, and 0.01; the two body classes lie geometrically between the head and tail instead of being assigned ad hoc counts. Binary targets use $n^*_{\text{head}}/n^*_{\text{tail}} = \rho_{\text{target}}$ directly. The same exponential rule is applied to every multiclass dataset–regime, so a given ρ_{target} denotes the same allocation shape in each of them.

The values n_i^* are real-valued targets, whereas a manifest must contain whole patches or slides. *Constrained integer allocation* converts them into usable class counts. Each target is rounded to the nearest integer and clipped to the class's support floor and available unique support. If the rounded counts do not sum to T, units are added to or removed from the classes with the largest rounding residuals until the total is exact. The resulting integers n_i therefore satisfy $\sum_i n_i = T$ and every class-specific lower and upper bound while remaining close to the exponential targets under this deterministic adjustment.

Counts determine how many units a class contributes; construction orderings determine which units they are. One master ordering is created per class at each hierarchy level. Every condition selects a deterministic prefix, or a split-unit-round-robin extension of a prefix, from these same orderings. For patch classification, all fixed-pool split units and slides contribute before additional patches are allocated; for MIL, slides are selected by cycling through the split-unit prefix before revisiting a unit.

If a target ratio is infeasible, the same rule is applied to every dataset and regime: use the largest attainable ratio and report the requested ratio, achieved ratio, limiting class, realized counts, and binding support floor. Validation or test support is never reduced to manufacture feasibility. This general fallback is especially relevant to high-K tasks, where an exponential profile must allocate support to many intermediate classes as well as the tail; it is not a dataset-specific exception. A stable lower-severity condition is preferred to a nominal $\rho = 100$ condition supported by only a few independent units.

The fallback has a limit: a moderate or severe condition whose realized allocation is indistinguishable from the balanced condition—achieved ratio within a small tolerance of 1, or realized counts identical to the balanced manifest—is not a lower-severity instance of the intended contrast but a null one, since the imbalance intervention did not occur. Such a condition is reported as degenerate rather than fitted and averaged with genuine moderate or severe conditions from other splits or assignments, and it is excluded from the imbalance deficit and recovery estimands in Section 5.5. Its requested ratio, achieved ratio, limiting class, and binding support floor are recorded alongside the exclusion, using the same fields every condition already reports.

The profile determines support by rank; the tail assignment maps semantic classes to those ranks. Native order is the clinical, ordinal, or natural-support order of the target; aligned and reversed order the classes by the difficulty scores d_c measured above. Binary tasks retain native and reversed orientations only. A difficulty mapping that duplicates an earlier mapping in any locked split is omitted across that dataset–regime with its reason recorded. These three assignments separate interpretable mechanisms but do not exhaust all possible class-support mappings. Figure 6 illustrates the design.

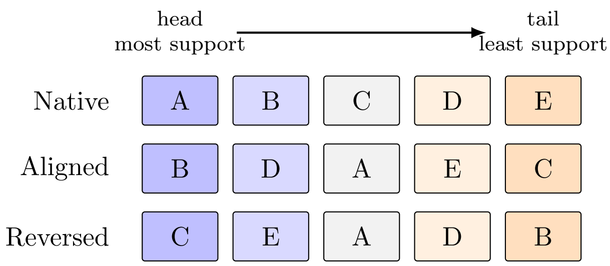

Figure 6: Illustrative tail assignments for a five-class target. Columns are support ranks fixed by the severity profile. Difficulty increases from B to C: aligned compounds difficulty with scarce support, while reversed compensates it.

The random quantities are separated into five seed families so that changing one source of variation does not silently change another. Table 6 defines their roles.

Seed family	Determines	Prespecified use
Split-unit partition	Membership of the training, validation, and test cohorts	Three locked patient- or case-disjoint splits
Tail assignment	Mapping of semantic classes to head–tail support ranks	Native, difficulty-aligned, and difficulty-reversed mappings from signed pilot evidence
Construction	Split-unit, slide, and patch orderings and the selected evidence	Three pilot orderings and a disjoint fourth ordering for definitive manifests
Initialization	Parameter initialization and optimization randomness	Two tuning seeds and five disjoint confirmation seeds
Resampling	Bootstrap draws and permutation swaps	Uncertainty intervals and hypothesis tests only

Table 6: Seed families and their non-overlapping roles in the benchmark.

No seed is chosen based on test performance. The size-matched balanced manifest and its trained CE baseline are shared by all tail assignments within a split.[2] The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

B Endpoint and Inference Specification

This appendix defines secondary endpoints, score semantics, resampling, and diagnostics. Gates and recovery retain the main-text definitions in Section 5.5.

B.1 Ordinal outcomes for PANDA

Let N be the number of held-out biopsies and C the number of grades. Let O_{ab} be the observed confusion-matrix count for true grade a and predicted grade b. The count expected under independent true and predicted grades is $E_{ab} = O_{a\cdot}O_{\cdot b}/N$, which retains the observed marginal grade frequencies. With $a, b \in \{0, \ldots, C-1\}$ and quadratic disagreement weight $w_{ab} = (a-b)^2/(C-1)^2$, quadratic-weighted kappa is

$$\kappa_{\mathrm{QW}} = 1 - \frac{\sum_{a=0}^{C-1}\sum_{b=0}^{C-1} w_{ab}O_{ab}}{\sum_{a=0}^{C-1}\sum_{b=0}^{C-1} w_{ab}E_{ab}}. \tag{5}$$

A value of 1 denotes perfect grade agreement, 0 is the agreement expected from the marginal grade frequencies, and negative values indicate more weighted disagreement than expected by chance. Ordinal mean absolute error (MAE) is

$$\mathrm{MAE}_{\mathrm{ord}} = \frac{1}{N}\sum_{i=1}^{N} |g_i - \widehat{g}_i|, \tag{6}$$

where g_i and $\widehat{g}_i$ are the true and predicted ISUP grades. It reports the average distance in grade levels and is lower-is-better. Kappa emphasizes agreement beyond the observed marginals, whereas MAE has the direct interpretation of grades missed per biopsy.

B.2 Probability quality and calibration

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \ldots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0,1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i.

Table 7 fixes which score variant enters each endpoint. For ordinary CE, post-hoc logit adjustment, and train-time logit adjustment, the target-prior variant is the correction defined in the companion methods report, using the natural-validation prior for the corresponding test split. For every other method, the target-prior variant equals its raw logits.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{7}$$

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i,y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from

[2] At $\rho_{\mathrm{target}} = 1$, every class receives equal support. Permuting semantic classes across support ranks therefore leaves the allocation unchanged, so tail assignment is not a meaningful factor for the balanced condition.

Endpoint or output	Score variant
Balanced accuracy and all other discrimination endpoints	Balanced-decision logits: $z^{\mathrm{CE}} - \tau \log \widehat{\pi}$ for post-hoc logit adjustment; raw model logits for train-time logit adjustment and every other method.
Tail-NLL deficit gate and tail-NLL recovery	Target-prior-corrected probabilities before temperature scaling.
Reported NLL, Brier score, ECE, and reliability diagrams	Target-prior-corrected probabilities before temperature scaling; raw probabilities are retained as a separately labelled diagnostic.
Temperature-scaled calibration outputs	Target-prior-corrected logits divided by one positive temperature fitted on natural-validation NLL and applied unchanged to test; these secondary outputs do not enter either recovery gate.

Table 7: Canonical mapping from evaluation endpoints to score variants.

dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\mathrm{NLL}_c = -\frac{1}{N_c} \sum_{i:y_i=c} \log \hat{p}_{ic}, \qquad \mathrm{NLL}_{\mathrm{macro}} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{NLL}_c, \tag{8}$$

where N_c is the number of held-out predictions from class c. Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2 \,. \tag{9}$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left| \frac{1}{|I_b|} \sum_{i \in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|} \sum_{i \in I_b} q_i \right|, \tag{10}$$

where I_b contains the predictions assigned to bin b. ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and split-unit weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. The output semantics and prior-correction formulas of individual methods are defined in the companion methods report (Queisler, 2026b).

B.3 Uncertainty from split-unit resampling

The resampled split unit is the patient or participant where one is available, and the biopsy/WSI for PANDA and CAMELYON16. Each replicate reweights the observed split units rather than deterministically repeating each one exactly once, and the complete statistic is recomputed from those weights.

Label-only preflight. Before model training, 10,000 bootstrap replicates are constructed using only split-unit identifiers, split membership, and class labels. *Label-only* means that no predictions, performance values, or method identities are used; these replicates test whether the planned resampling scheme is valid before results can influence it. The count of 10,000 is a prespecified Monte Carlo budget. It places about 250 replicates in each 2.5% tail of a percentile interval, reducing simulation noise in the interval endpoints while remaining inexpensive at this label-only stage.

Each split unit draws an independent, continuous bootstrap weight in every replicate: a Bayesian bootstrap (Rubin, 1981) in which the n observed units' weights are n times a draw from a symmetric Dirichlet$(1, \ldots, 1)$ distribution over the n-simplex. This draw is almost surely strictly positive in every coordinate, so every unit contributes to every replicate; no split–class cell, including one contributed by a unit whose observed split-by-class pattern is unique among the cohort, can be emptied by an unlucky draw. A unit's observed contribution pattern—how many examples it supplies to each class in each test-split appearance—is still recorded and grouped into strata, but only to report per-stratum diagnostics; it no longer constrains the weight draw itself. A selected unit's weight is applied to all of its slides and patches in every split. Figure 7 summarizes the workflow.

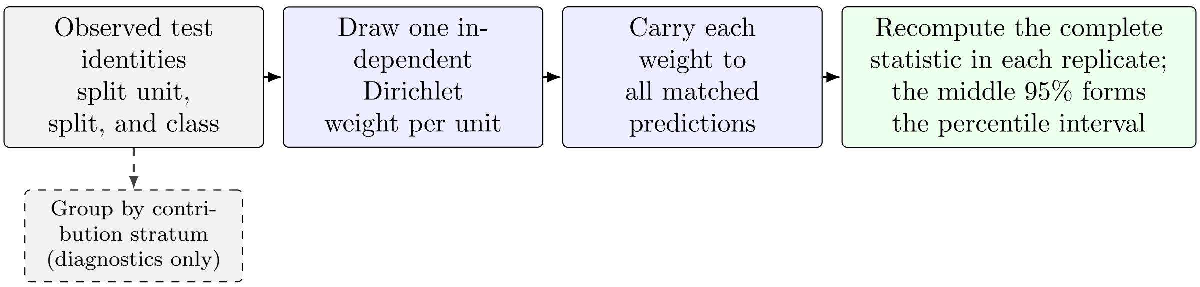

Figure 7: Split-unit-clustered Bayesian bootstrap workflow. The label-only preflight executes the identity, weight-draw, and stratum-diagnostic steps before predictions exist. After model fitting, the same paired per-unit weights are applied to every matched run.

The preflight verifies that every original split–class cell remains represented, every split-level metric is computable, one split unit receives the same weight across all of its split appearances, and every unit's resampled weight actually varies across replicates. Failure of any check stops the analysis rather than causing a replicate to be silently discarded.

The independent per-unit weight draw can still produce a replicate dominated by a few heavily weighted units. If split unit i has class-specific bootstrap weight v_i, the *Kish effective split-unit count*

$$n_{\text{Kish}} = \frac{\left(\sum_i v_i\right)^2}{\sum_i v_i^2}. \tag{11}$$

translates unequal case weights into the equivalent number of equally weighted split units. It decreases as weight concentrates. The preflight also records each split/class cell's number of contributing units and its largest unit-weight share. Table 8 lists the three concentration checks; failing any one designates inference for that dataset–regime descriptive before training.
A percentile rather than the minimum is used because the minimum across replicates is an extreme order statistic: its value falls as the replicate count rises, so a floor placed on it would encode the resampling budget rather than the support the cell actually carries. The diagnostic report and resampling seed are frozen with the analysis manifest.

Diagnostic	Threshold for descriptive-only designation
Effective split-unit count	2.5th percentile of n_{Kish} across replicates below five
Single-unit dominance	One unit supplies more than 50% of a class's weight in more than 5% of replicates
Cell support	A split/class cell has fewer than five contributing units

Table 8: Label-only preflight concentration checks. All three are evaluated before predictions exist; the designation is recorded rather than used to discard replicates.

Intervals after model fitting. The three split-unit partitions are fixed design repetitions and are averaged with equal weight; they are not treated as three independent cohorts. Tail assignments remain separate unless an assignment-averaged estimand is explicitly reported. Within each case-bootstrap replicate, the five matched confirmation-seed indices are also resampled as paired blocks and averaged before the splits are combined. This captures algorithmic variation without pretending that five model initializations create five additional case cohorts.

Balanced CE, imbalanced CE, and each mitigation method receive the same split-unit multiplicities and confirmation-seed draws. The complete deficit and recovery fraction, including numerator and denominator, is recomputed in each replicate. The reported 95% percentile interval spans the 2.5th to 97.5th percentiles of the resulting distribution. Patches are never bootstrapped independently.

C RQ3 Model Specification

This appendix fixes diagnostic feature construction and deliberately small association models used for RQ3. None of these quantities is allowed to use test outcomes as an input predictor.

Measuring separability and learnability. The two diagnostics differ only in their reference set: intrinsic separability uses one fixed balanced reference subset per dataset, regime, and split, whereas condition-specific learnability uses the exact balanced or imbalanced training manifest of the condition being explained. Both are evaluated on the unchanged natural-validation cohort. Balanced k-nearest-neighbour (kNN) macro recall labels each validation embedding from its closest reference embeddings and gives every class equal weight. A *linear probe* is a class-balanced linear classifier fitted on the frozen reference embeddings; its macro recall is the primary separability or learnability summary. Per-class nearest-neighbour error is reported descriptively to show which classes overlap. None of these diagnostics uses test labels or CE test performance. Patch probes operate on patch embeddings. WSI probes use the mean and componentwise standard deviation of a slide's eligible instances, avoiding a learned attention summary from the model being explained.

Sensitivity to correlated patches. Unique split-unit and slide counts remain the primary support measures. As a secondary sensitivity analysis, within-case redundancy is summarized by the intraclass correlation (ICC) of a cross-fitted class margin: an ICC near zero means that patches from the same case are no more alike than patches from different cases, and an ICC near one indicates strong redundancy.

For class c, let G_c be the number of split-unit or slide clusters, m_{gc} their sizes, and $N_c = \sum_g m_{gc}$. With between-cluster and within-cluster mean squares $\mathrm{MS}_{B,c}$ and $\mathrm{MS}_{W,c}$, the unequal-cluster-size correction and ICC are

$$m_{0,c} = \frac{N_c - \sum_{g=1}^{G_c} m_{gc}^2 / N_c}{G_c - 1}, \qquad \mathrm{ICC}_c = \frac{\mathrm{MS}_{B,c} - \mathrm{MS}_{W,c}}{\mathrm{MS}_{B,c} + (m_{0,c} - 1)\mathrm{MS}_{W,c}}. \tag{12}$$

The estimate is clipped to $[0, 1]$. The corresponding effective sample size is

$$N_{\mathrm{eff},c} = \frac{N_c}{1 + (\overline{m}_c - 1)\,\mathrm{ICC}_c}, \tag{13}$$

where $\overline{m}_c$ is the arithmetic mean cluster size. The scalar margin is the fixed intrinsic linear probe's logit for class c minus its largest competing logit. It is evaluated by *cross-fitting*: each observation is scored only by a probe trained without it. Equations (12)–(13) provide a patch-only sensitivity measure, not a replacement for independent-unit counts. Slide count is used directly for WSI targets, with patient count also examined when patients contribute multiple slides.

Three deliberately small association models. One model is fitted for each of the three RQ3 outcomes. The occurrence model handles the yes/no gate result; the magnitude and recovery models handle continuous outcomes. Every primary model contains exactly two predictors: the log achieved imbalance ratio and support–difficulty alignment A. The magnitude model also uses each deficit's bootstrap standard error so that a noisy estimated deficit is not treated as an exact observation.

Comparisons from the same dataset–target group can share a baseline level of difficulty. A *random intercept* allows each group to have its own starting level while estimating common associations with imbalance and alignment. This adjustment is important because the analysis pools several severities and assignments from the same group. The recovery model omits method-specific effects because the small number of gated units cannot support them.

The remaining predictors in Table 4 are fitted one at a time in separate sensitivity models. They are not combined with one another or with the two primary predictors because only a few independent dataset–target groups are available and the candidate predictors are strongly confounded. No interactions are fitted.

Parameters are estimated by maximum a posteriori (MAP) fitting: ordinary model fit is combined with weak zero-centered Gaussian priors that discourage extreme slopes and variance estimates in this small sample. A leave-one-dataset–target-group-out check then refits the magnitude model while withholding one complete group at a time. This is only a coarse stability check, not evidence of predictive generalization. TCGA-UT supplies the only genuine paired comparison between prediction regimes; the CAMELYON16, PANDA, and BRACS targets differ by regime. Model definitions, predictor roles, and priors are frozen before recovery outcomes are inspected, and all RQ3 conclusions remain hypothesis-generating.

D Training, Tuning, and Execution Specification

This appendix records exact update budgets, validation-search rules, replication, and resource accounting. Main-text interpretation depends only on their matched and test-blind use.

D.1 Update budgets and checkpoint selection

The frozen encoder, classifier family, optimizer family, batch definition, augmentation, and numerical precision are held fixed within a regime. Training is budgeted only in optimizer updates; epochs are neither a stopping rule nor a unit of comparison. With B the regime's locked batch size and T the common controlled support, the reference budget is

$$U = 30 \left\lceil \frac{T}{B} \right\rceil \tag{14}$$

updates, corresponding to 30 reference passes through T examples under ordinary sampling. Table 9 allocates that budget to the methods that do not consume it in a single stage.

Within a run, the reported checkpoint is fixed by three rules:

Method	Optimizer updates
Single-stage default	U; every method not listed below
RankMix-inspired	U teacher, then U student after reinitialization
cRT	U stage one, then $\lceil 0.2U \rceil$ stage two on the classifier only
MDE-inspired	U joint, each consuming its prespecified natural and balanced minibatch
Natural anchor (CE only)	Equation (14) at its own support; descriptive, enters no matched contrast

Table 9: Allocation of the reference update budget U. Balanced sampling, replacement, synthetic bags, and dual loaders do not change U. Processed examples, unique-example exposures, and updates are all reported.

1. Validation metrics are stored at fixed update intervals. The interval is scaled per dataset so the number of stored validation checkpoints is comparable across the widely differing controlled supports T, and the resolved value is recorded with every run.
2. The selected checkpoint maximizes natural-validation balanced accuracy; macro F1 and then lower NLL are deterministic tie-breakers.
3. Test predictions are produced once from that locked checkpoint.

D.2 Control grids and adaptive validation search

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity, and the search adapts within a prespecified envelope rather than committing to a single fixed grid. Every trainable method draws its learning rate from the common envelope

$$\eta \in \{1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}, 1 \times 10^{-2}\}. \tag{15}$$

Table 10 lists each method's imbalance-specific control. Focal loss, CE + soft F1, and CE + soft MCC draw their strength from a seven-value envelope searched through the same adaptive window as the learning rate; every other control is a fixed four-value grid, since its domain is naturally bounded. Both axes follow the search below, in which an edge winner is never accepted as an optimum: a bounded number of window shifts replaces a single fixed grid, at the cost of one new candidate per shift.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0, 0.5, 1, 1.5, 2, 4, 8\}$ envelope; active window starts at $\{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0, 0.25, 1, 4, 16, 64\}$ envelope; active window starts at $\{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
RankMix-inspired	beta shape $\alpha \in \{0.5, 1, 2, 4\}$
SC-MIL	temperature $\tau \in \{0.05, 0.1, 0.5, 1\}$
MDE-inspired	consistency weight $\lambda_{\text{con}} \in \{0, 0.1, 0.25, 0.5\}$

Table 10: Prespecified method-specific controls, crossed with the active window of the learning-rate envelope in Equation (15). Focal loss and the two soft-label hybrids search their listed envelope through an adaptive four-value window; every other row is a fixed four-value grid. cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately.

Adaptive window search, run once per method, dataset, prediction regime, and imbalance severity.

1. Initialize each active window to four adjacent envelope values: $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}$ for the learning rate, and the start listed in Table 10 for an adaptive strength axis.
2. Fit the active learning-rate window crossed with the method's strength window or fixed grid: at most 16 candidates in any one round. Candidates fitted in an earlier round are reused, never retrained.
3. If an axis's winning value is interior to its window, record the axis as *resolved* and stop searching it.
4. If the winner is an edge value and the envelope extends further in that direction, shift the window one position toward that edge and return to step 2. Only the single newly added candidate is fitted; the three overlapping candidates are reused.
5. If the winner is an edge value at the envelope's boundary, retain the boundary value and record the axis as *tuning-limited.*

Round bounds. At most three rounds on the learning-rate axis, three on focal loss's γ axis, and two on the soft-F1 and soft-MCC strength axis, each bounded by its envelope's length. A resolved or tuning-limited axis is never retrained in a later round.

For weighted CE, balanced sampling, focal loss, CE + soft F1, and CE + soft MCC, an imbalance-specific strength of zero is mathematically identical to plain CE at the same learning rate: it applies no class reweighting or sampling adjustment, reduces the focal term to the plain cross-entropy loss, or drops the auxiliary term entirely. That value is therefore excluded from the searched window, and its metrics are copied from CE's already-fitted same-learning-rate run rather than retrained, giving each of these five methods one additional comparison point at no extra training cost.

Four method families deviate from the search above. CE and methods without a meaningful imbalance-specific control search the learning-rate envelope alone. Train-time logit adjustment is tuned jointly over τ and learning rate, with τ held to its fixed grid. Post-hoc logit adjustment inherits the selected CE model and tunes only τ, because it performs no gradient optimization. cRT inherits the selected CE configuration for stage one and selects its stage-two classifier learning rate separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

D.3 Replication, locking, and records

Selection and replication follow prespecified units rather than per-run judgement:

- **Tuning seeds.** Every candidate is fitted with the same two tuning initialization seeds for each prespecified split-unit partition and tail assignment.
- **Selection unit.** One method, dataset, prediction regime, and imbalance severity. Within it, natural-validation balanced accuracy is averaged across both tuning seeds, all tuning splits, and all tail assignments; the highest aggregate is selected, with macro F1 and then lower NLL as deterministic tie-breakers.
- **Locking.** The chosen configuration is locked before any of the five disjoint confirmation initialization seeds is fitted and before test evaluation. Tuning runs never enter confirmation effect estimates.
- **Confirmation.** Each locked partition and tail assignment is repeated with the five confirmation seeds. Exactly three split-unit partition seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis.

- **Matching.** The two runs of a controlled contrast (Appendix E) share split-unit partition, tail assignment, confirmation-seed index, and held-out cases, so the contrast does not mix the intended change with a different cohort or replication.
- **Failures.** The unit of replication and all missing or failed runs are reported. A failed run is never silently replaced using a test-informed seed.

Split counts, support floors, condition manifests, tuning and confirmation seeds, method grids, and update budgets are frozen in the signed analysis manifest before the first definitive fit. Every candidate fitted, every window shift, and the resolved-or-tuning-limited outcome of each axis are recorded there before the corresponding method enters confirmation. No test data and no confirmation-stage result ever informs a shift, and no grid is changed once confirmation training begins.

For every realized condition, the experiment record contains

- dataset version and eligibility rules;
- split-unit, slide, and patch identifiers by partition;
- requested and achieved T and ρ, and the class assignment;
- all seed families and their tuning, confirmation, pilot, or definitive roles;
- pilot patch quota and bootstrap-preflight diagnostics;
- model and optimizer configuration, candidate grid, and selected checkpoint;
- environment identifier and test-prediction hash.

Deviations are dated and justified before the affected test labels are accessed.

Computational cost is separated into validation search and locked-configuration fitting, and reported as wall-clock training time, accelerator-hours, and peak accelerator memory; total and trainable parameter counts; and effective passes through unique training examples. Costs include every method-specific stage, including both cRT stages, but exclude the one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

E Terminology

Natural anchor. The full eligible natural training partition, fitted with the CE reference alone. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M.

Imbalance severity. The requested training allocation summarized by ρ; achieved severity is computed from realized integer counts.

Pilot level. A candidate number of independent training units per class used in the balanced CE support-stability study.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Comparison unit. One dataset, prediction regime, imbalance severity, and tail assignment evaluated together.

Deficit gate. A prespecified threshold rule that determines whether an observed discrimination or calibration deficit is large and precise enough for recovery analysis.

Imbalance deficit. The paired balanced-CE minus imbalanced-CE difference at common T and on the same held-out cases.

Recovery. A method's improvement over imbalanced CE divided by the evidenced imbalance deficit.

Kish effective split-unit count. The equally weighted split-unit count with the same weight concentration as one bootstrap replicate.

Intraclass correlation. The estimated similarity of class margins from patches belonging to the same split-unit or slide cluster.

Patch-micro estimate. A metric computed over eligible patch predictions, with split-unit-clustered uncertainty.

Case-macro estimate. An equal-weight aggregation over slides or split units, limiting domination by heavily tiled cases.

Self-contained search. The validation-search phase in which every candidate is fitted from scratch, needing no other method's result.

CE-inherited search. The validation-search phase for post-hoc logit adjustment and cRT, which reuse a condition's selected CE configuration and tune only what remains open for that method (τ for post-hoc logit adjustment; the stage-two learning rate for cRT), and so cannot begin until that condition's CE search has resolved.

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